Meng Lu

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Research Summary

Ph.D. researcher specializing in agentic LLM/VLM systems and efficient multimodal reasoning, with experience building scalable training environments, reward frameworks, and inference-time signals that help language and vision-language models reason, use tools, and adapt to real-world decision-making tasks. Thesis work focuses on maximizing productivity per token in LLM/VLM inference by jointly exploring algorithmic, systems, and hardware-level signals.

Education

Virginia Tech *Blacksburg, VA*
Doctor of Philosophy in Computer Science, GPA: 3.82/4.0 *Aug 2023 – Present*

- Advisor: Dr. Xuan Wang; Research Area: Agentic LLM/VLM systems, tool-integrated reasoning, efficient multimodal inference
- Graduate Research Assistant, Sanghani Center for Artificial Intelligence and Data Analytics

Northwestern University *Evanston, IL*
Master of Science in Computer Science, GPA: 3.7/4.0 *Sep 2013 – Mar 2015*

Hefei University of Technology *Hefei, China*
Bachelor of Engineering in Automation *Sep 2009 – Jun 2013*

Experience

Cisco AI Research *Oct 2025 – Present*
Research Intern, Agentic RL & Embodied AI *Los Angeles, CA*

- **Scaled agentic reinforcement learning** for tool-integrated VLM reasoning by integrating heterogeneous tool-calling actions into VLM agents, enabling dynamic tool selection, execution feedback, and multi-step planning across visual reasoning tasks.
- **Designed reward functions and embodied VLM-agent pipelines** that connect high-level visual-language planning with low-level robotic control for open-vocabulary manipulation, jointly evaluating tool-selection correctness, execution efficiency, and end-to-end task success.
- **Explored co-evolutionary training paradigms** where agent policy and task/environment distribution are jointly optimized to improve continual adaptation and out-of-distribution generalization.

Virginia Tech – Sanghani Center for AI and Data Analytics *Aug 2023 – Present*
Graduate Research Assistant, Advisor: Dr. Xuan Wang *Blacksburg, VA*

- **Led VISTA-Gym / Scaling Agentic RL for Tool-Integrated Reasoning in VLMs** (CVPR 2026, first author): built a scalable Gym-style training environment for VLM agents with diverse visual and symbolic tools, supporting reinforcement learning over multi-step tool use and reasoning trajectories; extended to MedVistaGym for tool-integrated medical visual reasoning.
- **Led multi-agent collaboration research** through CrossAgentIE, TriageAgent, and MAF-IE, developing cross-type, cross-task, and fine-tuned multi-agent frameworks for zero-shot IE and clinical triage.
- **Designed efficient VLM reasoning studies** through Think-Deep-VLM, analyzing layer-wise computational dynamics and inference-time selection signals across multimodal reasoning benchmarks.

Selected Publications

Meng Lu, Ligeng Zhu, Xuan Wang. “VICO: Visual Environments Co-Evolving for Vision-Language Model Reasoning”. NeurIPS 2026 submission

Meng Lu, Ran Xu, Yi Fang, Wenxuan Zhang, Yue Yu, Gaurav Srivastava, Yuchen Zhuang, Mohamed Elhoseiny, Charles Fleming, Carl Yang, Zhengzhong Tu, Yang Xie, Guanghua Xiao, Hanrui Wang, Di Jin, Wenqi Shi, Xuan Wang. *“Scaling Agentic Reinforcement Learning for Tool-Integrated Reasoning in VLMs.”* CVPR 2026. arXiv | GitHub

Meng Lu, Yuzhang Xie, Zhenyu Bi, Shuxiang Cao, Xuan Wang. *“CrossAgentIE: Cross-Type and Cross-Task Multi-Agent LLM Collaboration for Zero-Shot Information Extraction.”* ACL 2025. ACL | GitHub

Meng Lu, Brandon Ho, Dennis Ren, Xuan Wang. *“TriageAgent: Towards Better Multi-Agent Collaboration for LLM-Based Clinical Triage.”* EMNLP 2024. ACL | GitHub (Earlier version: ICML 2024 AI4Science Workshop Spotlight, 13% acceptance rate).

Meng Lu, Yuchen Zhuang, Wenqi Shi, Gaurav Srivastava, Charles Fleming, Xuan Wang. *“MAF-IE: Multi-Agent Finetuning for Zero-Shot Information Extraction.”* Under Review.

Meng Lu, Ligeng Zhu, Xuan Wang. *“Think-Deep-VLM: VLM-Native Internal Signals for Efficient Inference.”* Under

Review. (Collaboration with NVIDIA)

Meng Lu, Gaurav Srivastava, Xuan Wang. *"Towards Small (Vision-)Language Models as the Future of Real-World Agents."* Under Review.

Gaurav Srivastava, Aafiya Hussain, Zhenyu Bi, Swastik Roy, Priya Pitre, **Meng Lu**, Morteza Ziyadi, Xuan Wang. *"BeyondBench: Benchmark-Free Evaluation of Reasoning in Language Models."* ICLR 2026. arXiv

Gaurav Srivastava, Zhenyu Bi, **Meng Lu**, Xuan Wang. *"DEBATE, TRAIN, EVOLVE: Self-Evolution of Language Model Reasoning."* EMNLP 2025.

Zhenyu Bi, Gaurav Srivastava, Yang Li, Swastik Roy, **Meng Lu**, Morteza Ziyadi, Xuan Wang. *"JudgeBoard: Benchmarking and Enhancing Small Language Models for Reasoning Evaluation."* AAAI 2026. arXiv

Selected Projects

VISTA-Gym | *Python, PyTorch, Ray, Gymnasium, vLLM, DeepSpeed* *GitHub* | *arXiv*

- First-authored a **scalable agentic RL environment for tool-integrated visual reasoning**, unifying 7 reasoning tasks across 13 public VQA datasets with a standardized interface over 26 visual and symbolic tools, including Grounding DINO, EasyOCR, chart/document parsers, and math solvers.
- Built **high-throughput training infrastructure** with Ray/Gymnasium-compatible APIs, asynchronous rollouts, tool execution, trajectory logging, reward computation, and persistent VLM/tool microservices.
- Trained **VISTA-R1-8B**, an agentic VLM that outperforms comparable open-source VLMs by 9.51%–18.72% across reasoning-intensive VQA benchmarks.

MedVistaGym | *Python, PyTorch, Gymnasium, Medical VLMs, RL* *GitHub* | *arXiv*

- First-authored a **medical VLM training environment for tool-integrated reasoning over medical images**, supporting multi-step interaction with frozen visual tools such as segmentation, optical flow, temporal recognition, and object/tip detection models.
- Trained **MedVista-R1-8B**, which exceeds comparably sized tool-augmented baselines by 19.10%–24.21% across six medical VQA benchmarks, demonstrating that structured agentic training unlocks effective tool-integrated reasoning for medical image analysis.

Think-Deep-VLM | *Python, PyTorch, vLLM, Hugging Face* *Efficient VLM Reasoning*

- Developed a **training-free framework for analyzing layer-wise computational dynamics in VLMs**, supporting hidden-state projection into vocabulary space, divergence-based depth metrics, and inference-time response selection across Qwen3-VL and InternVL families.
- Identified a **systematic divergence between LLM and VLM internal reasoning signals**, showing that text-only depth-based internal signals do not directly transfer to multimodal reasoning settings.

Co-evolution VLM | *Python, PyTorch, vLLM, Hugging Face* *Self-Evolving VLM Training*

- Proposed a **co-evolution framework for visual-language models**, where the VLM and its training environment (tasks, tools, or data) are iteratively optimized to improve multi-step reasoning under limited supervision.
- Designed a **self-improving training loop** that jointly updates model policies and environment difficulty via automatic data generation, tool interaction feedback, and curriculum adaptation.
- Demonstrated **improved reasoning efficiency and generalization**, achieving stronger performance under low-resource or zero-shot settings compared to static training pipelines.

Honors & Awards

Sanghani Center Student Spotlight, Sanghani Center for AI and Data Analytics, Virginia Tech *2024*

ICML 2024 AI4Science Workshop Spotlight Paper, acceptance rate 13% *2024*

Technical Skills

Languages: Python, C/C++, Java, R, MATLAB, SQL

ML/RL Frameworks: PyTorch, Hugging Face Transformers, vLLM, DeepSpeed, Ray, Gymnasium, verl, LangChain

Models: Qwen-VL, InternVL, Llama, Mixtral, BERT/RoBERTa, vision-language and tool-augmented agents

Systems: distributed multi-GPU training/inference, RL rollouts, agent environments, Docker, AWS, SageMaker, FastAPI, Git